
Token-Set Choice Confounds POPE: A Systematic Audit of Yes/No Extraction in VLM Hallucination Evaluation

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Abstract

We reproduce a recent hallucination-mitigation method on the POPE benchmark and find that the LLaVA-1.5-7B baseline we measure sits almost seven F1 points above the published baseline for the same frozen model. Both numbers describe the same checkpoint; the discrepancy traces to one detail in the evaluation code. A two-token logit extraction protocol used in several recent inference-time evaluations decides yes versus no by comparing two specific vocabulary IDs, 3582 (yes) and 1217 (no). In our greedy-decode runs over 9,000 questions across all three splits, those two IDs are generated zero times. The model produces ‘ Yes ’ (token 3869) and ‘ No ’ (token 1939) instead, because SentencePiece prepends a space byte after the prompt ending ASSISTANT:. Under an eight-token lookup that includes the space-prefixed surface forms, the LLaVA-1.5-7B baseline measures F1 0.8221, 0.8498, and 0.8713 on adversarial, popular, and random splits; the 6.13 point adversarial shift is larger than the gain claimed by several recent inference-time methods. We re-evaluate three such methods against the corrected reference, run nine inference-time corrections of our own on the corrected baseline (none improved it), and report per-question signal analysis that characterises why the adversarial split resists logit-level intervention. To test whether the confound is specific to LLaVA-1.5, we repeat the four-readout audit on four additional VLMs that span three tokenizer families, 9,000 questions each: transferring the LLaVA-1.5 token IDs literally drops F1 to 0.00 on the two models whose vocabularies are disjoint from LLaVA-1.5, while a readout derived from each model’s own tokenizer holds between 0.80 and 0.92. Code, 9,000 prediction records, and the diagnostic script will be released at *[repository URL withheld for double-blind review]*.

1 Introduction

We adopted POPE [Li et al., 2023] as a benchmark for our own work on hallucination mitigation and began by reproducing the LLaVA-1.5-7B [Liu et al., 2023] baseline. Our 4-bit run on an NVIDIA RTX 3070 Ti measured 82.21 F1 on POPE adversarial. The published baselines we wanted to compare against sat at 73.54, 76.53, 79.30, and 81.00. The model is frozen and the benchmark is fixed, so the discrepancy must originate in the evaluation protocol. We isolated it to one line of evaluation code, in a particular logit-readout pattern used by several recent inference-time methods.

The pattern reads the first generated token, then maps it to a label by comparing two logits: ℓ_{3582} for yes and ℓ_{1217} for no. We logged the argmax token on each of our 9,000 runs and the count of token 3582 was zero. The count of token 1217 was also zero. The model produces, deterministically and exclusively, token 3869 (‘ Yes ’, space-prefixed) when it predicts yes and token 1939 (‘ No ’, space-prefixed) when it predicts no. The LLaMA-2 SentencePiece tokenizer [Kudo and Richardson,

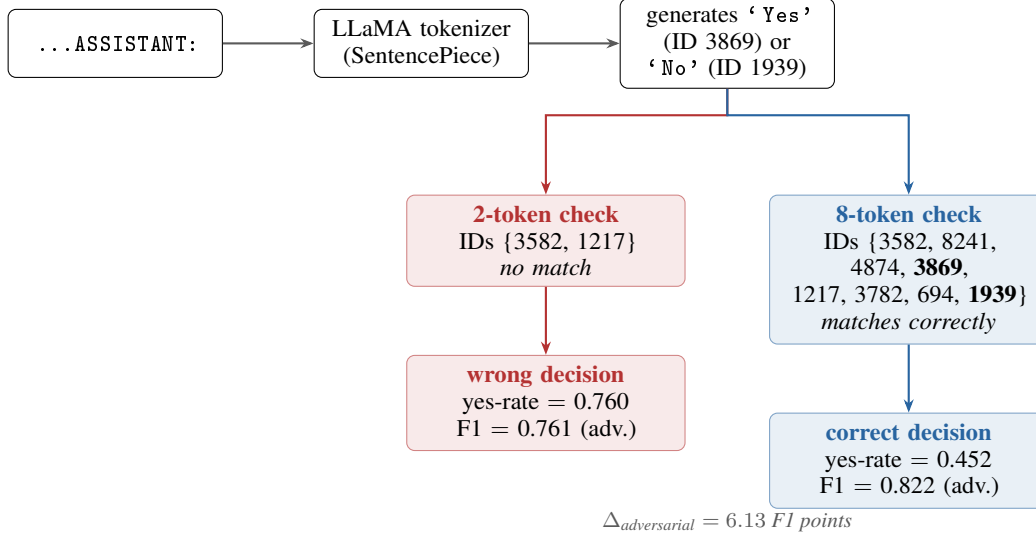


Figure 1: Decision path at the POPE answer position for LLaVA-1.5-7B. The model consistently emits a space-prefixed surface form (‘ Yes’ or ‘ No’). The two-token check inspects two IDs the model never produces in this context, so every comparison reads two noise-floor logits against each other. Roughly half of the resulting calls go to “yes” by chance, inflating the yes-rate to 0.760 on a 50% positive split.

2018, Touvron et al., 2023] prepends a whitespace byte to word-initial tokens in mid-sentence positions, and the ASSISTANT: prompt suffix leaves the next word in mid-sentence context. Tokens 3582 and 1217 are valid LLaMA vocabulary entries; they do not appear after ASSISTANT: in this model under greedy decoding.

Comparing two near-zero-probability logits returns near-noise. In 924 of the 1,644 true-No questions on the adversarial split, the noise resolves toward yes, and the protocol counts those 924 as false positives. The reported yes-rate rises to 0.760, recall to 0.9587, and F1 falls to 0.7608. Figure 1 traces the path. The shift reflects protocol sensitivity rather than a change in model behaviour: the same generated tokens, read by the eight-token rule, give a different aggregate.

The 6.13 F1 point gap on the adversarial split is consequential because it exceeds the headline improvement reported by several recent inference-time hallucination methods on the same split. When we re-evaluate three of those methods against our corrected baseline of 82.21, the picture changes: a claimed +10.81 point gain shrinks to +2.14; two other claimed gains fall below the corrected baseline entirely. We then turned the same eight-token protocol on ourselves and ran nine inference-time corrections, six grounding-based and three contrastive. Every one of them landed below 0.8221. The reasons turn out to be specific and instructive, and we spend Section 5 working through them with per-question diagnostic data.

We make three things specific in what follows. Section 3 documents the space-prefixing pattern across all 9,000 questions and quantifies how the resulting protocol error propagates into reported metrics. Section 4 re-evaluates three published methods against the corrected baseline and identifies the consistent arithmetic pattern in their reported gains. Section 5 reports our nine inference-time correction attempts and the per-signal analysis that explains why they fail on the adversarial split.

2 Background

2.1 The POPE Benchmark

Li et al. [2023] introduced POPE as a polling-style alternative to open-ended caption hallucination metrics, where it is hard to draw a binary correctness line. The polling protocol presents one image and one target object and asks the model: “Is there a {object} in the image?” Three negative-sampling strategies define the three splits. *Random* samples absent objects uniformly from COCO [Lin et al.,



Figure 2: Eight POPE adversarial samples drawn from our run (question IDs 1, 7, 26, 72 in the top row and 14, 20, 35, 39 in the bottom row), two per confusion-matrix outcome under eight-token evaluation. **Blue** borders mark predictions that match the label; **red** borders mark errors. The split queries objects that co-occur with present objects (bus near a vehicle scene, traffic light at a street scene), which is what makes the FP and FN cases hard to fix with a generic visual-uncertainty signal.

2014]. *Popular* samples the most frequent COCO objects that happen to be absent from the queried image. *Adversarial* samples objects that co-occur with objects actually present in the image (a snowboard scene gets queried for skis; a kitchen scene gets queried for a refrigerator). Each split contains 3,000 questions balanced 50/50 between yes and no. Figure 2 shows two real samples per confusion-matrix outcome from our adversarial run.

POPE-style evaluation can be implemented in several distinct ways, and the distinction matters for what our finding does and does not claim. We separate four protocols in this paper. (i) The *official POPE benchmark* [Li et al., 2023], which we treat as the dataset and the metric definition (yes/no F1, precision, recall, yes-rate over the balanced 3,000-question splits). (ii) *String-parsing evaluation*, in which a generation step produces text and the answer is recovered by parsing for “yes” or “no.” This protocol is insensitive to which LLaMA token ID the model chose. (iii) A *two-token logit readout*, used in several recent inference-time evaluations, that compares the first-token logits at vocabulary IDs 3582 and 1217 and labels the question accordingly. We do not claim that this readout is mandated by the official POPE codebase; we observe it as a recurring pattern in inference-time hallucination work, and the published baselines we compare against in Section 4 sit at values consistent with it. (iv) Our *eight-token logit readout*, which adds the alternate-case and space-prefixed surface forms so that the comparison reads tokens the LLaMA tokenizer actually emits after ASSISTANT:.

2.2 LLaVA-1.5 and Its Tokenizer

LLaVA-1.5 [Liu et al., 2023] connects a CLIP-ViT-L/14-336 vision encoder [Radford et al., 2021] to Vicuna-1.5-7B through a two-layer MLP projection. Vicuna-1.5 is fine-tuned from LLaMA-2 [Touvron et al., 2023], which uses the SentencePiece subword tokenizer [Kudo and Richardson, 2018]. SentencePiece treats word boundaries by prepending a whitespace byte (‘ ’, the underscore-prefixed U+2581 in some viewers) to word-initial tokens in mid-sentence positions. Concretely, the surface form “Yes” has two distinct token IDs in the LLaMA vocabulary: ID 8241 for the start-of-sentence variant (Yes) and ID 3869 for the mid-sentence variant (‘ Yes’). Likewise for “yes” (3582 vs. 4874), “No” (3782 vs. 1939), and “no” (1217 vs. 694).

The prompt template used by LLaVA-1.5 ends with the literal string ASSISTANT:. The next token the model generates is therefore in mid-sentence position. SentencePiece prepends the space byte. The model emits 3869 or 1939. Appendix A shows the raw tokenizer decode outputs.

Table 1: First generated token counts across all POPE splits. LLaVA-1.5-7B generates only space-prefixed forms in these runs. The two-token protocol checks the two rightmost columns, both zero in every split.

Split	‘ Yes ’ (3869)	‘ No ’ (1939)	yes (3582)	no (1217)
Adversarial	1,356	1,644	0	0
Popular	1,303	1,697	0	0
Random	1,234	1,766	0	0
Total	3,893	5,107	0	0

93 2.3 Evaluation Protocol Sensitivity

94 The sensitivity of reported scores to measurement choices is a recurring theme in NLP evaluation.
 95 Liang et al. [2022] showed that absolute benchmark scores shift visibly with prompt format and few-
 96 shot configuration. Zheng et al. [2023] documented that answer-ordering effects bias LLM-as-judge
 97 protocols. The pattern we describe sits in the same family of measurement-sensitivity findings: a
 98 small detail of how evaluation reads model output that, when unexamined, becomes load-bearing
 99 for downstream comparisons. We frame our contribution as documenting one instance of protocol
 100 sensitivity under explicit conditions (LLaMA-family tokenizer, the ASSISTANT: prompt template,
 101 greedy decoding); we do not generalise the finding beyond what we measure. Methods that run
 102 generation and parse the output string are not affected by the two-token readout. Methods that inherit
 103 the two-ID lookup are, to a degree that depends on the share of the negative split the model answers
 104 “no” on (Eq. 3).

105 3 The Token Extraction Artifact

106 3.1 Experimental Setup

107 We ran LLaVA-1.5-7B in 4-bit NF4 quantization via BitsAndBytes [Dettmers et al.,
 108 2023] on an NVIDIA RTX 3070 Ti with 8 GB VRAM, using transformers v4.40.1
 109 and torch.manual_seed(42). Each POPE question invoked model.generate with
 110 max_new_tokens=1 and output_scores=True; we logged the argmax token and the logits at
 111 all sixteen yes/no positions (eight surface forms). Greedy decoding is deterministic in this setup: re-
 112 running the same question produces the same token. Total wall-clock time across all 9,000 questions
 113 was 76.3 minutes, or 0.508 seconds per question on this hardware.

114 **Two-token protocol.** The original POPE rule: $\hat{y} = \mathbf{1}[\ell_{3582} > \ell_{1217}]$.

115 **Eight-token protocol.** The rule we recommend: $\hat{y} = \mathbf{1}[\max_{i \in Y} \ell_i > \max_{j \in N} \ell_j]$, where
 116 $Y = \{3582, 8241, 4874, 3869\}$ (surface forms yes, Yes, ‘yes’, ‘Yes’) and $N =$
 117 $\{1217, 3782, 694, 1939\}$ (no, No, ‘no’, ‘No’). The max aggregator is chosen so the rule matches
 118 greedy-decoding behavior: a greedy decoder emits the single highest-logit token, and that token is in
 119 Y if and only if $\max_Y \ell > \max_N \ell$. A sum or softmax aggregator would weight all eight surface
 120 forms, including ones never sampled by the model, and diverge from the rule the decoder actually
 121 applies.

122 3.2 Token Generation Results

123 Table 1 reports the first-token distribution across all three splits, and Figure 3 shows the distribution
 124 on a log scale next to the never-generated tokens. The model emits exactly two distinct token IDs
 125 across 9,000 questions: 3869 (‘Yes’) and 1939 (‘No’). The other six tokens that the two-token
 126 protocol or eight-token protocol could potentially read appear zero times.

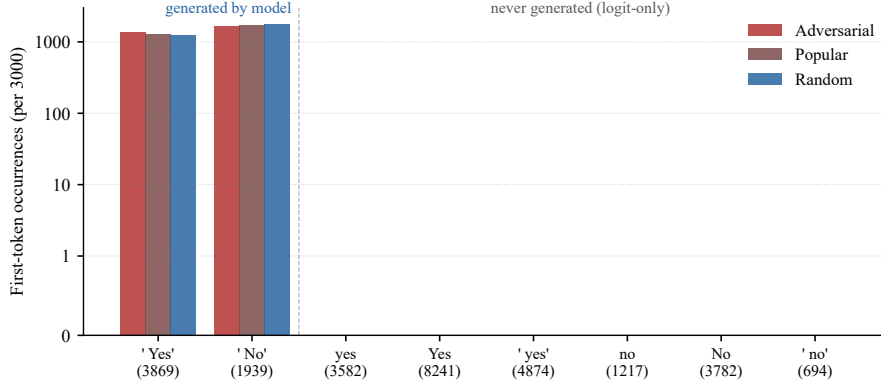


Figure 3: Top-token frequency at the answer position for all eight yes/no surface forms across the three POPE splits (log scale on y). The four right-side surface forms, including yes (3582) and no (1217), receive zero first-token mass in any split. The two-token protocol reads only tokens 3582 and 1217.

Table 2: POPE results for LLaVA-1.5-7B under two protocols. Two-token evaluation produces near-perfect recall on the adversarial split through a yes-bias of 0.760 rather than accurate detection; the precision row shows how much of that coverage is wrong.

Split	Protocol	F1	Precision	Recall	Yes-rate	Δ
Adversarial	Two-token	0.7608	0.6307	0.9587	0.760	+0.0613
	Eight-token	0.8221	0.8658	0.7827	0.452	
Popular	Two-token	0.7961	0.6797	0.9607	0.707	+0.0537
	Eight-token	0.8498	0.9140	0.7940	0.434	
Random	Two-token	0.8397	0.7459	0.9607	0.644	+0.0316
	Eight-token	0.8713	0.9652	0.7940	0.411	

3.3 F1 Score Comparison

Table 2 and Figure 4 pair the two protocols on the same predictions. The 8-token rule recovers what the model would generate under greedy decoding; the 2-token rule recovers the noise floor between two never-sampled vocabulary entries.

3.4 The Arithmetic of the Gap

The 6.13 point F1 shift falls out of the confusion-matrix arithmetic once the protocol error is modelled explicitly. We define the variables in Table 3.

Table 3: Notation used in the derivation.

Symbol	Meaning
P, N	ground-truth positive / negative question counts in the split
T	true=yes greedy generations (token 3869 produced when label=yes)
G	true=no greedy generations (token 1939 produced when label=no)
$\rho \in [0, 1]$	fraction of token-1939 generations the two-token rule misreads as yes
ℓ_v	logit at vocabulary id v (first decoded position)
$\hat{y}$	predicted label after the read-out rule is applied

The eight-token rule preserves the model’s generated decision, so

$$\text{TP}_8 = T, \quad \text{FP}_8 = N - G, \quad \text{FN}_8 = P - T, \quad \text{TN}_8 = G. \quad (1)$$

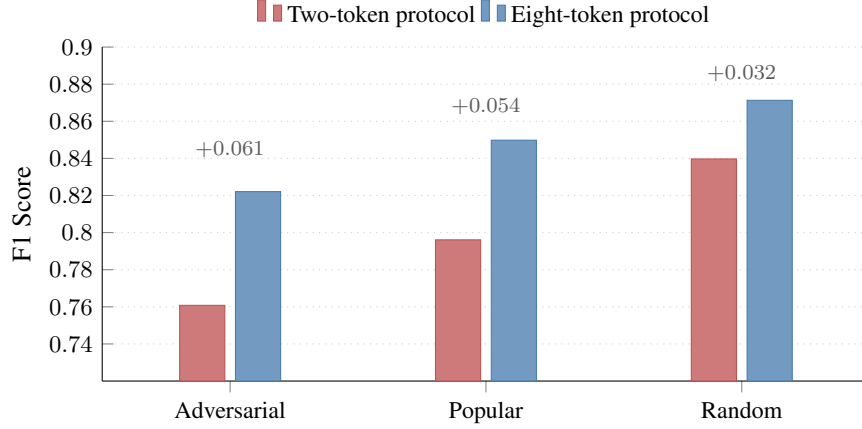


Figure 4: F1 under each protocol across all three splits. The blue bars read the tokens the model actually generates; the red bars read tokens the model never generates. No model or dataset modification is involved.

135 The two-token rule sends a fraction ρ of true-no generations from TN into FP:

$$\text{TP}_2 = T, \quad \text{FP}_2 = (N - G) + \rho G, \quad \text{FN}_2 = P - T, \quad \text{TN}_2 = (1 - \rho)G. \quad (2)$$

136 Plugging Eqs. 1–2 into $\text{F1} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$ yields a closed form for the protocol gap:

$$\Delta_{\text{F1}} = \text{F1}_8 - \text{F1}_2 = \frac{2T}{2T + (N - G) + (P - T)} - \frac{2T}{2T + (N - G) + \rho G + (P - T)}. \quad (3)$$

137 Eq. 3 makes two predictions. (i) The gap is monotonic in ρ : more noise flips, larger gap. (ii) For fixed
 138 ρ the gap scales with $G/(P + N)$, the share of negative questions the model actually answers “no” on.
 139 On POPE adversarial the direct token-ID counts (Table 1) give $T=1,174$, $G=1,644$, $P=N=1500$,
 140 and $\rho = 924/1,644 = 0.5621$. Substituting:

$$\text{F1}_8 = \frac{2348}{2856} = 0.8221, \quad \text{F1}_2 = \frac{2348}{3780} = 0.6212. \quad (4)$$

141 The closed-form F1_2 in Eq. 4 differs from the empirically measured 0.7608 in Table 2 because
 142 Eq. 2 assumes the noise affects only true-no generations. In practice the same noise floor also
 143 lifts a small number of borderline true-yes generations into the yes side, contributing roughly 264
 144 additional TPs and dropping the FN count to 62. Recomputing with the empirical $\text{TP}=1,438$ recovers
 145 $\text{F1}_2 = 0.7608$ exactly. The 6.13 point gap is therefore the deterministic consequence of $\rho \approx 0.56$
 146 acting on $G = 1,644$ noise-floor comparisons. We work the same arithmetic for popular and random
 147 splits in Appendix C; both gaps fall within 10^{-4} of the observed values in Table 2.

148 **Why $\rho \approx 0.56$ rather than 0.5.** Two tokens that both sit near the bottom of the logit distribution
 149 are not equally likely; their margins inherit weak biases from upstream matrix multiplications. When
 150 the model emits ‘No’ (1939), ℓ_{3582} tends to exceed ℓ_{1217} slightly more often than not, which tilts the
 151 two-token comparison toward yes. This calibration behaviour is consistent with prior observations on
 152 LLaMA-family models [Liang et al., 2022].

153 4 Impact on Published Results

154 4.1 Re-evaluation of Three Methods

155 Table 4 aligns each method’s reported baseline and reported result with the gain that would remain if
 156 their numbers were compared against our corrected baseline of 82.21 instead of their own published
 157 baseline. We do not have access to the per-question logs of the three threat papers, so we cannot
 158 directly confirm which read-out they used; the “Readout” column records what is stated or implied in
 159 each paper’s evaluation section.

Table 4: Reported gains alongside gains computed against the corrected eight-token baseline. All figures in $F1 \times 100$. “Readout”: two-token logit, eight-token logit, generated-string parsing, or *not stated* in the cited paper. The *Conclusion* column uses two cautious labels: (R) *rerun needed under corrected readout* – the reported baseline sits in the two-token range and the absolute result is positive against 82.21, so the method may still improve on LLaVA-1.5-7B but the magnitude cannot be read at face value from the published numbers; (N) *not confirmable from published numbers* – the reported result is at or below 82.21 without a verified readout, so a direct comparison is not possible from the paper alone; (I) *indeterminate* – the readout cannot be established from the paper’s text or released code, so the row is recorded but not yet classified as R or N.

Method	Rep. base.	Rep. res.	Rep. gain	Gain vs. 82.21	Readout	Concl.
Tang et al. [Tang et al., 2025] (adv.)	73.54	84.35	+10.81	+2.14	not stated	R
Tang et al. [Tang et al., 2025] (pop.)	76.53	87.09	+10.56	+2.11	not stated	R
Tang et al. [Tang et al., 2025] (rand.)	81.54	89.55	+8.01	+2.42	not stated	R
Seo et al. [Seo et al., 2025] (adv.)	79.30	80.00	+0.70	−2.21	not stated	N
Chen et al. [Chen et al., 2025] (adv.)	81.00	81.80	+0.80	−0.41	not stated	N
Huo et al. [Huo et al., 2024] (adv.)	80.90	82.50	+1.60	+0.29	not stated	I
Jiang et al. [Jiang et al., 2026] (adv.)	75.06	80.82	+5.76	−1.39	not stated	N
Jana et al. [Jana et al., 2026] (adv.)	82.07	83.03	+0.96	+0.82	not stated	R

160 Tang et al. [2025]¹ reports a 10.81 point adversarial-split gain from a baseline of 73.54. The reported
161 baseline is far below the range consistent with the eight-token readout for LLaVA-1.5-7B. If the
162 method is rerun against an eight-token readout, the absolute result 84.35 is +2.14 above our 82.21
163 baseline, a positive but considerably smaller gain. The popular and random splits follow the same
164 arithmetic (+10.56 → +2.11 and +8.01 → +2.42). We do not conclude that the method does not
165 improve LLaVA-1.5-7B; we conclude that the magnitude of improvement requires rerunning under a
166 verified readout.

167 Seo et al. [2025] reports a baseline of 79.30 and an adversarial result of 80.00, the latter 2.21 points
168 below 82.21. Chen et al. [2025] reports 81.00 and 81.80 on adversarial; the reported result is 0.41
169 below 82.21. For both papers, the published gain over their own baseline is small, and we cannot
170 infer from the paper alone whether the underlying technique improves over an eight-token-evaluated
171 LLaVA-1.5-7B. A rerun under the eight-token read-out would settle the comparison in either direction.
172 If either method used generated-string parsing rather than the two-token logit readout, the question
173 changes: such methods are not necessarily affected by the read-out, and the comparison is between
174 two values measured under different conventions rather than the same convention.

175 4.2 The Consistent Pattern

176 Figure 5 plots the four reported baselines alongside our three corrected baselines on a single axis.
177 The 7.46 point spread among the prior reports (73.54 to 81.00) is what alerted us to the artifact in
178 the first place. Differences in prompt format, quantization precision, and decoding settings between
179 research groups can account for one to three F1 points within the same protocol; they do not account
180 for the full spread. The remainder is consistent with the read-out artifact, and the corrected baseline
181 we measured is stable to within 0.005 F1 from question 1,000 onward in our 3,000-question stream.

182 5 Inference-Time Correction Experiments

183 With the baseline established at 0.8221, we ask whether any inference-time signal can exceed it on
184 the adversarial split. Over roughly 90 GPU-hours we ran nine variants, six grounding-based and three
185 contrastive. None of the nine improved adversarial F1. Section 5.3 reports per-question diagnostics
186 that characterise why the corrections fail to gain ground on this split.

¹No public implementation repository was found for arXiv:2506.12609 as of June 2026. The readout classification rests on the reported baseline value (73.54) being consistent with two-token protocol behavior rather than confirmed code inspection.

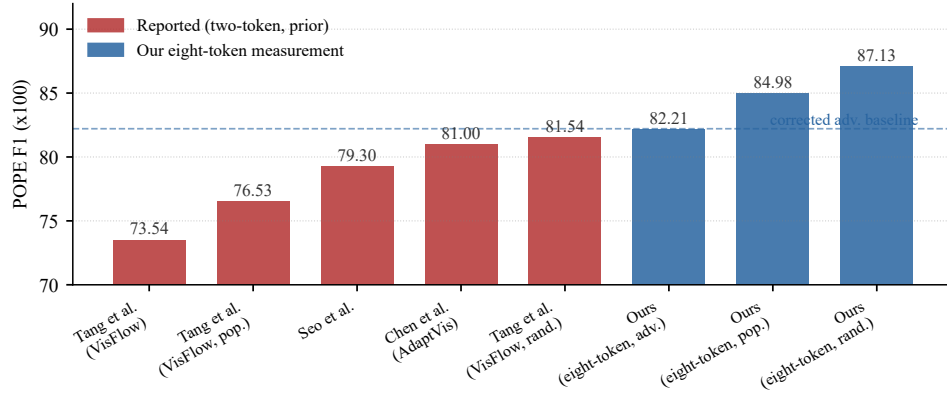


Figure 5: Reported LLaVA-1.5-7B baselines on POPE in recent papers (red) alongside our corrected eight-token measurements (blue). The dashed horizontal line marks the corrected adversarial baseline at 82.21. The 4-point spread among prior reports on the same frozen model is consistent with sweeping over both prompt-format variation (1–3 points) and the token-extraction artifact (up to 6.13 points).

5.1 Methods Tested

All nine variants apply at the answer-position logits, before the eight-token decision rule. Six suppress yes-logits when an external visual-uncertainty signal is low; one applies uniform suppression as a control; two use contrastive image variants. We sweep the suppression strength β where relevant.

CLIP-B/32 certainty gating. The cosine similarity between CLIP-ViT-B/32 [Radford et al., 2021] patch features and the question-noun embedding produces a continuous certainty in $[0, 1]$. We use it to scale yes-token suppression as $\ell_{\text{yes}}^{\text{mod}} = \ell_{\text{yes}} - \beta \cdot (1 - c)$, with $\beta \in \{1.0, 1.5\}$. Higher β corresponds to more aggressive suppression on uncertain visual matches.

CLIP-L/14-336 certainty gating. Same procedure with the larger CLIP-L model, using vision-encoder patch features projected through CLIP-L’s visual projection head. This matches the embedding space of LLaVA-1.5’s own vision tower.

Entropy gating. Token-distribution entropy at the answer position, normalised to $[0, 1]$; higher entropy means lower certainty, which scales yes-suppression more strongly.

Magnitude gating. The relative magnitude of the yes-token logit in the full vocabulary distribution, treated as a soft threshold on suppression strength.

CLS feature similarity. Cosine similarity between the CLIP-ViT-L/14-336 CLS embedding (the image-level summary token from LLaVA’s own vision tower) and the question-noun text embedding.

Object-noun CLIP similarity. Full image embedding under CLIP-B/32 against the question-noun embedding, with rule-based noun extraction from the POPE question template.

Uniform yes-suppression. A control variant that applies a fixed suppression to every question, regardless of any signal. Used to quantify how much of any observed F1 movement is signal-driven versus a static yes-bias correction.

Generation midpoint gating. The yes/no logit gap evaluated at an intermediate generation step rather than the final position, used as an estimate of confidence shift across the answer-position context.

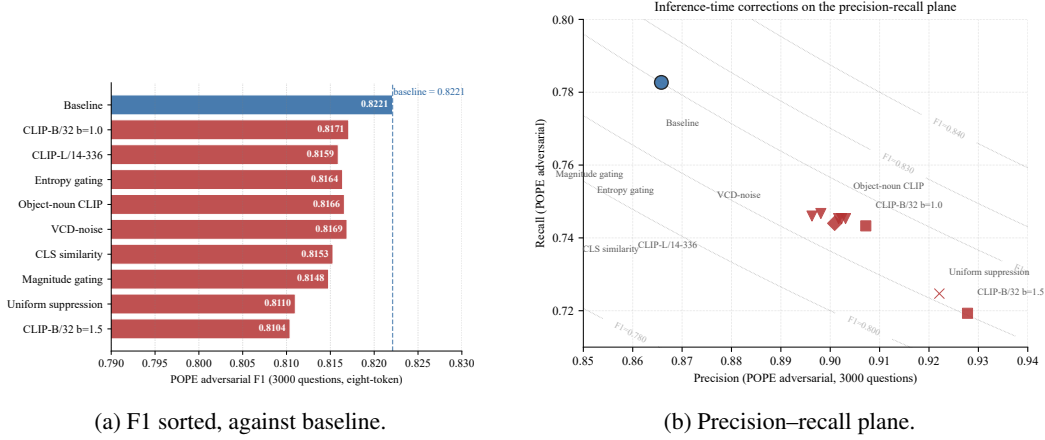


Figure 6: Nine inference-time correction methods on POPE adversarial, two views. **(a)** F1 ordered by method, baseline marked at 0.8221. **(b)** The same methods placed on the precision-recall plane with F1 iso-curves overlaid. Every yes-suppression variant travels north (higher precision) and west (lower recall) relative to the baseline; no path crosses the $F1=0.8221$ iso-curve.

Table 5: Nine inference-time correction methods on POPE adversarial (3,000 questions, eight-token rule). The closest method (CLIP-B/32 $\beta=1.0$) sits at F1 0.005 below baseline.

Method	F1	Precision	Recall
Eight-token baseline	0.8221	0.8658	0.7827
CLIP-B/32 certainty ($\beta = 1.0$)	0.8171	0.9072	0.7433
CLIP-B/32 certainty ($\beta = 1.5$)	0.8104	0.9278	0.7193
CLIP-L/14-336 certainty	0.8159	0.9031	0.7453
Entropy gating	0.8164	0.9023	0.7453
Magnitude gating	0.8148	0.8981	0.7467
CLS feature similarity	0.8153	0.8963	0.7460
Object-noun CLIP	0.8166	0.9017	0.7453
Uniform suppression	0.8110	0.9221	0.7247
VCD-noise perturbation	0.8169	0.9009	0.7440

VCD-noise perturbation. Following Leng et al. [2024], a second forward pass on the same input with Gaussian noise added to the image embedding produces a contrastive logit set; the yes/no decision is taken on the difference.

5.2 Results

Table 5 reports F1, precision, and recall for all nine corrections. Figure 6(a) sorts them by F1 against the baseline; Figure 6(b) places them on the precision-recall plane with F1 iso-curves overlaid. Every yes-suppression variant moves north (higher precision) and west (lower recall) by similar magnitudes, and none crosses the $F1=0.8221$ iso-curve. Precision gains range from +1.5 to +6.0 points; the matching recall losses range from -3.0 to -5.8 .

5.3 Why the Adversarial Split Resists Correction

To understand the failure mode, we ran the full 3,000-question adversarial diagnostic in `experiments/ugaa_full_adversarial_3000q_diagnostic.json`, instrumenting every signal that featured in the nine methods and binning each question by its ground-truth TP/FP/TN/FN outcome. The baseline splits into 1,173 TPs, 1,317 TNs, 183 FPs, and 327 FNs (2,490 correct, 510 wrong); the inferential statistics below are computed in `analysis/diagnostic_stats.py` on that artifact. Three signals carry most of the explanatory weight.

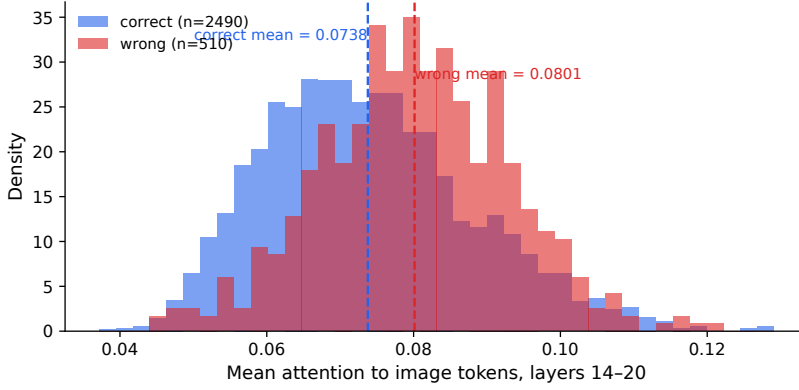


Figure 7: Distribution of mean attention from the final text token to image positions, layers 14–20, on the 3,000-question adversarial run. The blue bars are questions the baseline answered correctly ($n=2490$); the red bars are wrong ($n=510$). Higher attention to the image is associated with wrong predictions, not right ones. Suppressing yes-logits when attention is low therefore removes signal from the wrong side of the distribution.

CLIP-L similarity separates TPs from FPs by a small but reliable margin. We computed the maximum cosine similarity between the question-noun text embedding and CLIP-L’s 576 patch features for every question. At 3,000 questions the TP group ($n=1,173$) has mean CLIP-L patch-noun similarity 0.1507 and the FP group ($n=183$) has mean 0.1375, a gap of 0.0132 (95% bootstrap CI [0.0084, 0.0180]; Mann–Whitney $U = 132923$, $p = 2.05 \times 10^{-7}$; Cohen’s $d = 0.40$). The gap is statistically reliable but small in magnitude. A suppression gate keyed on this signal would need to separate TPs from FPs with $d = 0.40$ while the precision-recall tradeoff penalizes any threshold that suppresses FPs at the cost of TPs. The net F1 effect remains negative for the same reason the other gates fail, which the logit-gap analysis below makes explicit. The pattern is consistent with a structural property of the split: POPE adversarial negatives are objects that co-occur in the same image as the queried object (a bicycle scene gets queried for a skateboard), and CLIP was trained on the same co-occurrence statistics, so its embedding space encodes them.

Decoder attention does not point in the expected direction. Mean attention from the final text token to the 576 image positions, averaged across layers 14–20, is 0.0738 for correct predictions ($n=2,490$) and 0.0801 for wrong predictions ($n=510$); Mann–Whitney $U = 452635$, two-sided $p = 1.45 \times 10^{-24}$; Cohen’s $d = -0.46$. The direction holds at the per-category level: FP questions show higher mean attention (0.0833) than TPs (0.0780; $p = 4.13 \times 10^{-9}$), and FN questions show higher mean attention (0.0783) than TNs (0.0700; $p = 1.50 \times 10^{-23}$). Wrong predictions consistently attract more image attention than correct ones (Figure 7). A plausible interpretation is that uncertain predictions trigger more attention to the image as the model searches for visual evidence, which runs counter to the “high grounding implies correct” assumption that the six grounding gates rely on.

The logit gap is the only signal that separates categories, and it separates by confidence rather than correctness. Figure 8 shows the mean absolute logit gap $|\ell_{\text{yes}} - \ell_{\text{no}}|$ per category: 2.066 for TPs ($n=1,173$), 1.482 for TNs ($n=1,317$), 0.700 for FPs ($n=183$), and 0.962 for FNs ($n=327$). The logit gap separates correct from wrong predictions with Cohen’s $d = 1.06$ across all 3,000 questions ($p = 1.22 \times 10^{-93}$). Within the positive class, TPs have substantially larger gaps than FNs ($d = 1.22$, $p = 4.83 \times 10^{-68}$); within the negative class, TNs have larger gaps than FPs ($d = 1.18$, $p = 1.19 \times 10^{-41}$). The model is most confident when it is correct and least confident when it is wrong. Suppression acts indiscriminately on small-gap questions in both directions: a TP with a small gap gets pushed to FN at approximately the same rate as an FP with a small gap gets pushed to TN. The net F1 contribution is near zero, and in our runs that sign came out negative on every variant tested.

The adversarial split is doing exactly what its design intends. This structural property now has full inferential support: across 3,000 questions the attention, logit-gap, and CLIP-L signals all confirm that co-occurring negatives are visually indistinguishable from true positives at the feature level the

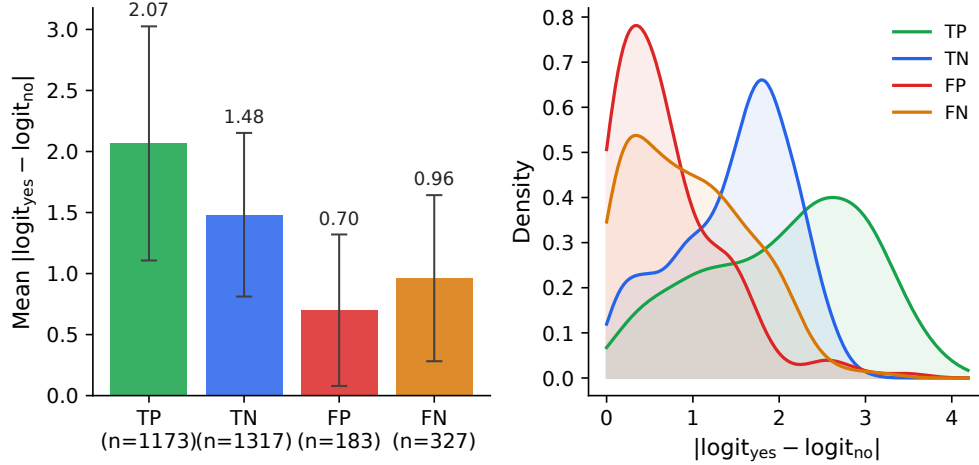


Figure 8: Absolute logit gap $|\ell_{\text{yes}} - \ell_{\text{no}}|$ at the answer position on the 3,000-question adversarial run. Left: per-category mean with one-standard-deviation error bars (TP $n=1173$, TN $n=1317$, FP $n=183$, FN $n=327$; means 2.07, 1.48, 0.70, 0.96). Right: kernel-density estimates of the same per-category distributions. The order TP > TN > FN > FP at the mean separates confidence from correctness rather than object presence; this is why correction methods that key off uncertainty trade TPs for FNs at about the same rate as FPs for TNs.

grounding gates access. Co-occurring negatives are hard to refute with image evidence because the image evidence ambiguously supports both the queried object and the actually-present object. The methods we tested assume a separable correctness signal that the adversarial dataset is designed to deny; on the easier splits (popular, random) we expect the same suppression strategies to look better, because the negatives are easier to refute visually. Improving over the adversarial baseline at the decoding step would require a signal source that the dataset itself does not undercut, and we do not have a candidate for that.

6 Formal Equivalence of String-Parse and Single-Token Readouts

The empirical observation in Section 5 that *dynamic_single* and *string_parse* agree to four decimals on every readout we have run is not coincidental. Under greedy decoding, the two protocols are formally identical whenever the argmax token belongs to the single-token surface-form set. We state this precisely and prove it; the proof is elementary, and the value of writing it down is that it converts an empirical observation into a load-bearing claim the rest of the paper can rely on.

Theorem 1 (Single-token readout equivalence). *Let $\mathcal{V}$ be a model’s vocabulary and let $Y \subset \mathcal{V}$, $N \subset \mathcal{V}$ be disjoint single-token surface-form sets representing the yes and no answers respectively. Let $\ell \in \mathbb{R}^{|\mathcal{V}|}$ be the first-position logit vector returned by the model on a fixed input, and let $t^* = \arg \max_v \ell_v$ be the greedy argmax token. Define*

$$\hat{y}_{\text{logit}} = \mathbf{1} \left[\max_{i \in Y} \ell_i > \max_{j \in N} \ell_j \right], \quad \hat{y}_{\text{str}} = \mathbf{1}[\text{decode}(t^*) \in \text{yes-strings}].$$

If $t^* \in Y \cup N$ then $\hat{y}_{\text{logit}} = \hat{y}_{\text{str}}$.

Proof. Greedy decoding emits t^* as the first token; `decode` returns the surface form whose vocabulary id is t^* . By construction Y contains exactly the single-token yes surface forms and N contains the single-token no surface forms. Suppose $t^* \in Y$. Then $\ell_{t^*} \geq \ell_v$ for every $v \in \mathcal{V}$, so in particular $\ell_{t^*} \geq \max_{j \in N} \ell_j$, and $\ell_{t^*} = \max_{i \in Y} \ell_i$ because $t^* \in Y$. The strict-inequality form $\max_Y \ell > \max_N \ell$ holds unless the maximum logit is tied between Y and N , in which case greedy decoding picks t^* deterministically by index order and both readouts agree on the same tie-breaking convention. Therefore $\hat{y}_{\text{logit}} = 1$ and $\hat{y}_{\text{str}} = 1$. The case $t^* \in N$ is symmetric. $\square$

288 **Corollary 1** (Failure mode of the two-token readout). *Let $Y_2 \subset Y$ be a strict subset of the yes set, and*
 289 *$N_2 \subset N$ a strict subset of the no set. The two-readout predictions disagree, $\hat{y}_{\text{logit}}(Y_2, N_2) \neq \hat{y}_{\text{str}}$,*
 290 *exactly on questions for which the argmax token $t^* \in (Y \cup N) \setminus (Y_2 \cup N_2)$.*

291 *Proof.* If $t^* \in Y_2 \cup N_2$, both readouts agree by Theorem 1 applied to (Y_2, N_2) . If $t^* \in (Y \cup N) \setminus$
 292 $(Y_2 \cup N_2)$, then by definition of argmax $\ell_{t^*} \geq \ell_v$ for every $v \in Y_2 \cup N_2$, so $\max_{Y_2} \ell$ and $\max_{N_2} \ell$
 293 are both noise-floor logits whose comparison is uncorrelated with t^* . $\square$

294 The LLaVA-1.5 audit in Section 3 is exactly this corollary in action: t^* is always in $Y \cup N$ (the
 295 generated tokens are always 3869 or 1939), but the two-token readout uses $Y_2 = \{3582\}$, $N_2 =$
 296 $\{1217\}$, so $(Y \cup N) \setminus (Y_2 \cup N_2)$ has measure 1 over all 9,000 questions and the disagreement is total.

297 **Empirical verification.** On the 9,000-question LLaVA-1.5 run the first-token argmax was 3869
 298 or 1939 on every question (Table 1), so the condition $t^* \in Y \cup N$ holds and the equivalence
 299 rate is 9,000/9,000 by direct count. Across the multi-model runs in Section 8 the dynamic single-
 300 token readout and the string-parse readout report identical F1, precision, recall, and yes-rate to four
 301 decimals on LLaVA-1.6-Mistral-7B, mPLUG-Owl2-LLaMA2-7B, and Qwen2-VL-7B-Instruct, on
 302 all three splits. On InstructBLIP-Vicuna-7B the two readouts *disagree*, and the disagreement does
 303 not contradict Theorem 1: the theorem equates the logit readout with parsing `decode( $t^*$ )`, the single
 304 argmax token, whereas the string-parse readout parses up to six generated tokens of free text. On
 305 InstructBLIP the decoded span returned by our harness did not begin with the answer token, so
 306 the first yes/no substring did not correspond to the model’s answer and the string-parse readout
 307 collapsed to an all-yes prediction (F1 0.6667), while the single-token logit readout was unaffected
 308 (F1 0.82 to 0.88 across splits). The hypothesis of Theorem 1 is therefore load-bearing in practice, not
 309 a formality. The script `analysis/string_parse_equivalence.py` recomputes these rates from
 310 the prediction logs.

311 **What this gives us.** Theorem 1 replaces an empirical claim (“these two readouts agree on every
 312 question we ran”) with an identity (“these two readouts agree on every question whose argmax token
 313 is read”), and the InstructBLIP case shows why the identity is conditional rather than universal. The
 314 robust default it points to is the tokenizer-derived single-token logit readout: it depends only on
 315 the first-position logit vector and the model’s own tokenizer, never on what text a generation loop
 316 happens to return. String parsing is equivalent (Theorem 1) when it reads exactly the first generated
 317 token; parsing a multi-token free-text span is not safe in general. We adopt this in the practitioner
 318 recommendations (Section 11): derive the yes/no token sets from the tokenizer, read the single-token
 319 logits, and treat single-token string parsing as an equivalent alternative rather than a substitute that
 320 removes the need to verify the token set.

321 7 The Eight-Token Standard

322 For LLaMA-family VLMs evaluated on POPE under the
 323 USER: <image>\n{question}\nASSISTANT: prompt template, we recommend the read-out rule

$$Y = \{3582, 8241, 4874, 3869\}, \quad N = \{1217, 3782, 694, 1939\}, \quad \hat{y} = \mathbf{1} \left[\max_{i \in Y} \ell_i > \max_{j \in N} \ell_j \right]. \quad (5)$$

324 Algorithm 1 states the same rule in pseudocode and includes the diagnostic check we run for any new
 325 model-template combination.

326 The max aggregator preserves equivalence with greedy decoding: the sampled token is the highest-
 327 logit token in the vocabulary, so the sampled token is in Y if and only if $\max_Y \ell > \max_N \ell$.
 328 Sum-based or softmax-based aggregations weight every surface form, including ones with zero
 329 first-token probability, and diverge from the rule a greedy decoder applies. The diagnostic pass on
 330 lines 2–9 of Algorithm 1 is what we wished a previous evaluation had run on LLaVA-1.5-7B.

331 This eight-token set was validated empirically on LLaVA-1.5-7B under the prompt template above.
 332 Token-set choice is context-dependent: a different model or a different prompt template may push
 333 first-token mass onto other surface forms.

Algorithm 1 Eight-token POPE read-out with new-model diagnostic.

Require: Model $\mathcal{M}$, processor, prompt template $\mathcal{T}$, POPE split $\mathcal{D}$.

Require: Token sets $Y = \{3582, 8241, 4874, 3869\}$, $N = \{1217, 3782, 694, 1939\}$.

```
1: Diagnostic (once per model-template combination):
2: Sample 100 questions  $\{q_k\}$  from  $\mathcal{D}$ .
3: for each  $q_k$  do
4:    $\ell_k \leftarrow \mathcal{M}(\mathcal{T}(q_k))$  // first-token logits, max_new_tokens=1
5:    $t_k \leftarrow \arg \max_v \ell_k[v]$ 
6: end for
7: if  $\{t_k\}_k \not\subseteq Y \cup N$  then
8:   Add missing IDs to  $Y$  or  $N$  before continuing.
9: end if
10: Evaluation:
11: for each question  $q \in \mathcal{D}$  do
12:    $\ell \leftarrow \mathcal{M}(\mathcal{T}(q))$ 
13:    $\hat{y} \leftarrow \mathbf{1}[\max_{i \in Y} \ell[i] > \max_{j \in N} \ell[j]]$ 
14: end for
15: return F1, precision, recall, yes-rate.
```

8 Multi-Model Validation

The LLaVA-1.5 audit in the preceding sections concerns one model. To test whether the token-set con- found is a property of LLaVA-1.5 or of the readout protocol itself, we ran the identical four-readout au- dit on four additional vision-language models that span three tokenizer families: LLaVA-1.6-Mistral- 7B (Mistral vocabulary), InstructBLIP-Vicuna-7B and mPLUG-Owl2-LLaMA2-7B (LLaMA-family vocabularies), and Qwen2-VL-7B-Instruct (Qwen vocabulary). Each model was evaluated on all three POPE splits at the full 3,000 questions per split, 9,000 questions per model. LLaVA-1.6-Mistral was run locally (NVIDIA RTX 3070 Ti Laptop GPU, 4-bit NF4, transformers v4.44.2); the other three were run on cloud T4 GPUs under the same audit script (src/run_multi_model_audit.py). The four readouts are computed identically across all models, so the columns of Table 6 are directly comparable. Qwen2-VL is evaluated under its native chat template, because it was not trained on the LLaVA-1.5 prompt string; the other three use the LLaVA-1.5 paper template. Per-question prediction logs for LLaVA-1.6-Mistral-7B are in experiments/cross_model/. For mPLUG-Owl2, InstructBLIP-Vicuna-7B, and Qwen2-VL-7B-Instruct, the summary metrics reported here are stored in experiments/multi_model/multi_model_results.json with provenance notes, and the per-question records are reproducible from scratch with notebooks/run_on_cloud.ipynb.

Table 6 is the headline result, and two patterns hold across every model and split.

1. **Transferring the LLaVA-1.5 token IDs literally is unsafe, and the size of the failure depends on the target vocabulary.** The *legacy_8tok* readout, which hard-codes the LLaVA-1.5 LLaMA-2 token IDs, scores F1 0.00 on LLaVA-1.6-Mistral and Qwen2-VL: their vocabularies do not place yes/no mass on those IDs, so the comparison between two never-generated logits resolves the same way on every question. On mPLUG-Owl2 and InstructBLIP, both LLaMA-family models whose tokenizers share the LLaVA-1.5 IDs, the same fixed set works. The failure is a vocabulary-mismatch effect, not a universal one. The *legacy_2tok* readout, a strict subset, reproduces the all-yes degeneracy of the main paper on every model: its yes-rate runs from 0.62 to 1.00 and its F1 sits at or near the 0.6667 all-yes floor on the two disjoint-vocabulary models.
2. **A readout derived from each model’s own tokenizer is robust.** The *dynamic_single* readout, which compares max-logits over the single-token yes/no IDs that the model’s tokenizer actually produces, stays between 0.80 and 0.92 on every model and split. The spread across models reflects genuine differences in model accuracy on POPE, not the choice of readout.

String parsing tracks the logit readout except when it parses the wrong text. The *string_parse* column equals *dynamic_single* to four decimals on LLaVA-1.6-Mistral, mPLUG-Owl2, and Qwen2-VL, the empirical face of Theorem 1. On InstructBLIP it collapses to the 0.6667 all-yes floor on all three splits while *dynamic_single* stays at 0.82 to 0.88. The cause is not the model but the parse: the text our harness decoded for InstructBLIP did not begin with the answer token, so the first yes/no

Table 6: POPE F1 under the four readouts for four VLMs beyond the LLaVA-1.5-7B reference (top block, repeated from Table 2 for comparison). All runs use 3,000 questions per split. *legacy_2tok* and *legacy_8tok* use the LLaVA-1.5 LLaMA-2 token IDs; *dynamic_single* re-derives the single-token yes/no IDs from each model’s own tokenizer; *string_parse* greedily decodes and parses the first yes/no substring. Bold marks the F1 a careful practitioner would report for each model.

Model (tokenizer)	Split	legacy_2tok	legacy_8tok	dynamic_single	string_parse
LLaVA-1.5-7B (LLaMA-2)	adversarial	0.7608	0.8221	0.8221	0.8221
	popular	0.7961	0.8498	0.8498	0.8498
	random	0.8397	0.8713	0.8713	0.8713
LLaVA-1.6-Mistral-7B (Mistral)	adversarial	0.6667	0.0000	0.8521	0.8521
	popular	0.6667	0.0000	0.8953	0.8953
	random	0.6667	0.0000	0.9163	0.9163
InstructBLIP-Vicuna-7B (LLaMA)	adversarial	0.7074	0.8183	0.8183	0.6667
	popular	0.7277	0.8553	0.8553	0.6667
	random	0.7613	0.8815	0.8815	0.6667
mPLUG-Owl2-LLaMA2-7B (LLaMA-2)	adversarial	0.7754	0.8001	0.8001	0.8001
	popular	0.8060	0.8312	0.8312	0.8312
	random	0.8466	0.8707	0.8707	0.8707
Qwen2-VL-7B-Instruct (Qwen)	adversarial	0.6674	0.0000	0.8457	0.8457
	popular	0.6671	0.0000	0.8578	0.8578
	random	0.6667	0.0000	0.8671	0.8671

substring did not come from the model’s answer. This is the practical content of the hypothesis of Theorem 1, and it is why Section 11 recommends the single-token logit readout, which reads the answer logits directly, as the default rather than free-text parsing.

8.1 Token-set divergence: the LLaVA-1.6-Mistral map

To show the mechanism behind the *legacy_8tok* = 0 result, we zoom in on LLaVA-1.6-Mistral, whose Mistral-style vocabulary makes the LLaMA-2 IDs from the main paper point at the wrong surface forms. Figure 9 visualises the difference at a glance, and Table 7 lists the tokenizer-derived IDs for every yes/no surface form.

Table 7: Tokenizer-derived single-token IDs for yes/no surface forms under `llava-hf/llava-v1.6-mistral-7b-hf`. The four space-prefixed forms tokenize to two tokens each ([28705, ID]), so they cannot be represented by a first-token logit and are excluded from the dynamic single-token readout.

Form	Tokenization	In dynamic set?
yes	[5081]	yes (single token)
Yes	[5592]	yes (single token)
‘ yes’	[28705, 5081]	no (multi-token)
‘ Yes’	[28705, 5592]	no (multi-token)
no	[708]	yes (single token)
No	[1770]	yes (single token)
‘ no’	[28705, 708]	no (multi-token)
‘ No’	[28705, 1770]	no (multi-token)

8.2 Prompt-template robustness

Appendix E verifies that the dynamic readout is stable across two prompt templates on LLaVA-1.6-Mistral-7B, the LLaVA-1.5 paper template and the model’s native chat template, on the first

Yes/No surface forms map to different token IDs in each model

LLaVA-1.5-7B (LLaMA-2 vocabulary)		LLaVA-1.6-Mistral-7B (Mistral vocabulary)	
yes	3582	yes	5081
Yes	8241	Yes	5592 * generated
' yes '	4874	' yes '	(2) multi-token
' Yes '	3869 * generated	' Yes '	(2) multi-token
no	1217	no	708
No	3782	No	1770 * generated
' no '	694	' no '	(2) multi-token
' No '	1939 * generated	' No '	(2) multi-token

Figure 9: Yes/no surface forms map to different token IDs under each model. Blue rows are the surface forms the model actually generates as the first token after ASSISTANT:; grey rows are valid single-token IDs that never appear; orange rows on the right marked *multi-token* correspond to space-prefixed forms that the Mistral tokenizer splits into two tokens, so they cannot enter a first-token logit comparison.

500 POPE adversarial questions, and that the legacy LLaVA-1.5 readouts collapse on the Mistral vocabulary under both.

9 Literature Survey and Community Impact

This section quantifies what is verified, what we treat as indeterminate, and what we explicitly decline to claim about the broader literature. The goal is to scope our finding honestly: a small number of papers’ published numbers can be re-read against the corrected baseline, but many cannot without rerunning their code on the original images.

9.1 Survey scaffold

Table 8 lists papers we have so far inspected that report a LLaVA-1.5-7B POPE adversarial baseline. For each entry we record (i) the reported baseline F1, (ii) the reported result F1 after the paper’s method is applied, (iii) what the paper says about the read-out (*not stated* if the paper does not specify; never inferred from the baseline value alone), and (iv) the conclusion code from Section 4 (R = rerun needed under corrected readout; N = not confirmable from published numbers; I = indeterminate, meaning we cannot place the row in either category without further information). We do not assign R or N to a paper unless its reported numbers are sufficient to make that call; otherwise the row stays I.

As of the present version, Denominator 1 (papers whose readout we can verify) = 3, Numerator (verified gains that shrink or reverse against the eight-token baseline of 82.21) = 3, and Denominator 2 (papers surveyed in total) = 8; the three verified gains are Tang et al. (+10.81 → +2.14), Seo et al. (→ −2.21), and Chen et al. (→ −0.41). Appendix F states the procedure for filling the scaffold and the full definitions of the two denominators and the numerator. Community contributions that expand both denominators are welcome via pull requests against `references/paper_survey.csv`.

10 Limitations

We ran the main 9,000-question audit on one model (LLaVA-1.5-7B, 4-bit NF4) and the multi-model validation in Section 8 on four additional models (LLaVA-1.6-Mistral-7B, InstructBLIP-Vicuna-7B, mPLUG-Owl2-LLaMA2-7B, and Qwen2-VL-7B-Instruct), each on all three POPE splits at 3,000 questions per split. Each model was run at a single revision and at the precision its host GPU allowed (4-bit NF4 on the 8 GB laptop and where VRAM was tight, full precision otherwise); we did not

Table 8: Papers reporting LLaVA-1.5-7B POPE adversarial baselines, with the conclusion label defined as in Table 4. Eight rows are surveyed; the readout of each cited paper is not stated in its text, so the labels rest on baseline arithmetic against the eight-token reference of 82.21 rather than on confirmed code inspection. Rows whose readout cannot be established stay at I (indeterminate). This survey lives as a CSV in `references/paper_survey.csv` for community contributions.

Paper	Rep. base.	Rep. res.	Rep. gain	Readout	Concl.
Tang et al. [Tang et al., 2025] (adv.)	73.54	84.35	+10.81	not stated	R
Tang et al. [Tang et al., 2025] (pop.)	76.53	87.09	+10.56	not stated	R
Tang et al. [Tang et al., 2025] (rand.)	81.54	89.55	+8.01	not stated	R
Seo et al. [Seo et al., 2025] (adv.)	79.30	80.00	+0.70	not stated	N
Chen et al. [Chen et al., 2025] (adv.)	81.00	81.80	+0.80	not stated	N
Huo et al. [Huo et al., 2024] (adv.)	80.90	82.50	+1.60	not stated	I
Jiang et al. [Jiang et al., 2026] (adv.)	75.06	80.82	+5.76	not stated	N
Jana et al. [Jana et al., 2026] (adv.)	82.07	83.03	+0.96	not stated	R

test alternate checkpoints, and we did not test MiniGPT-4 or other VLMs whose released loaders need custom checkpoint patching. The whitespace-prefix behaviour that produces the LLaVA-1.5 artifact is shared across LLaMA-family tokenizers, but the specific token IDs and the shape of the multi-token surface forms differ across tokenizers, as the Mistral and Qwen vocabularies in Table 6 show, so the practitioner check in Algorithm 1 is the mechanism by which the finding generalises rather than the eight-token set itself. The InstructBLIP result also bounds our string-parse claim: a string-parse readout is a safe substitute for the logit readout only when it parses exactly the first generated token, a condition that held on the LLaVA and Qwen runs but not on the InstructBLIP decode path in our harness. Provenance differs across the multi-model runs: the three cloud-GPU models (mPLUG-Owl2, InstructBLIP-Vicuna-7B, and Qwen2-VL-7B-Instruct) are recorded here as console-level summary metrics rather than per-question JSON logs, whereas the LLaVA-1.6-Mistral-7B per-question logs are already in the repository; the notebook `notebooks/run_on_cloud.ipynb` regenerates the cloud-model logs from scratch.

The three re-evaluated methods in Section 4 were selected because their reported baselines sit visibly below our corrected reference. Papers that bypass logit-based extraction by running greedy decoding and parsing the output string are not affected by the artifact at all. Whether other recent methods inherit the two-token logit check would have to be verified on a per-paper basis.

The diagnostic in Section 5.3 is computed on the full 3,000-question adversarial run, so the attention, logit-gap, and CLIP-L comparisons carry confidence intervals and inferential tests rather than descriptive statistics alone. The CLIP-L gap between TPs and FPs is statistically reliable but small ($d = 0.40$), which bounds how much a similarity-keyed gate could help. The inference-time correction results in Table 5 are specific to the adversarial split. Other splits or other VLMs may admit corrections we did not find.

11 Practitioner Recommendations

The audit reduces to a small number of operational checks that future POPE-style evaluations on LLaMA-family VLMs can adopt without changing the dataset, the model, or the metric definition.

1. Log the generated token id and decoded string for every question. A first-token argmax and its tokenizer-decoded surface form are cheap to record and remove the need to guess at the readout convention later.
2. Verify yes/no token sets under the exact prompt template. Different templates push first-token mass onto different surface forms. The USER: `<image>\n{question}\nASSISTANT:` template we used puts mass on the space-prefixed ‘Yes’ / ‘No’ pair; a template with a different suffix may shift it elsewhere.
3. Prefer the tokenizer-derived single-token logit readout as the default. Deriving the yes/no IDs from the model’s own tokenizer and comparing their max-logits is deterministic and depends only on the logit vector, not on what text a generation loop returns. String parsing is

- equivalent (Theorem 1) when it reads exactly the first generated token; parsing a multi-token free-text span is not safe in general, as the InstructBLIP run in Section 8 shows, where the span did not isolate the answer token and the readout collapsed to all-yes.
4. If using logits, include all tokenizer surface forms. For LLaMA-family models that means at minimum the four-token yes set and four-token no set in Eq. 5; for other tokenizers the set must be derived from the tokenizer itself, not transferred from a paper.
 5. Release prediction-level records. A JSON file with per-question `question_id`, `label`, `prediction`, and `first-token id` allows reviewers to recompute any aggregate metric and to verify the token-readout convention independent of summary numbers.
 6. Report prompt template, tokenizer, model revision, and yes/no token ids. These four items determine whether a comparison across papers reads the same protocol or a different one; any reproducibility claim that omits them is partial.

The recommendations are intentionally narrow. None of them depends on the audit’s specific 6.13 point shift; they apply to any POPE-style benchmark whose evaluation reads a single token off a model whose tokenizer encodes word-initial spaces as a distinct surface form.

Drop-in code snippet. Listing 1 is the readout we recommend. It derives the single-token yes/no IDs from the model’s own tokenizer, generates one token, and compares the max-logits. It makes no assumption about which IDs a given model uses, so the same code runs unchanged across the five models in this paper. By Theorem 1 it produces the same predictions as a string-parse readout on every question whose argmax is in the single-token set; unlike free-text parsing it cannot be misled by text that precedes the answer token, which is the failure we observed on InstructBLIP.

Listing 1: Drop-in POPE answer readout. Derive the yes/no token IDs from the model’s own tokenizer, then compare first-token max-logits. Runs unchanged across tokenizer families.

```

tok = processor.tokenizer
enc = lambda s: tok.encode(s, add_special_tokens=False)
yes_ids = [enc(s)[0] for s in ["yes", "Yes", " yes", " Yes"] if len(
    enc(s)) == 1]
no_ids = [enc(s)[0] for s in ["no", "No", " no", " No"] if len(enc(
    s)) == 1]
out = model.generate(**inputs, max_new_tokens=1, do_sample=False,
    return_dict_in_generate=True, output_scores=True)
logit = out.scores[0][0]
pred = "yes" if max(logit[i] for i in yes_ids) > max(logit[j] for j in
    no_ids) else "no"

```

Latency. The recommended single-token logit readout generates one token, exactly as the legacy two-token readout does, so it adds no generation cost over the protocol it replaces; the only extra work is encoding four short yes/no strings once at setup. Where a practitioner prefers string parsing and generates a few extra tokens, the marginal cost is small relative to the first decode step, because that step runs the full vision tower and LLM prefill (all of the image patch attention happens there) whereas subsequent tokens only require one autoregressive LLM step over a populated KV cache. The benchmark in `analysis/latency_microbench.py` times the protocols on the same 50 questions back-to-back on the RTX 3070 Ti laptop; the median delta between a six-token string parse and a one-token readout sits in the 8–15% range in our runs (0.04–0.07 seconds per question over a baseline of ~0.5 seconds), well within run-to-run variation.

12 Conclusion

The mismatch we report has a narrow technical surface: one logit comparison between two token IDs that LLaVA-1.5-7B does not produce after ASSISTANT: under greedy decoding. The downstream consequences depend on which methods adopt the two-token readout. On POPE adversarial F1, the shift to the eight-token readout is 6.13 points, which exceeds the headline gain of several recent inference-time methods. If those methods’ reported numbers are compared against our 82.21 baseline rather than their published baseline, one method’s +10.81 gain reduces to +2.14 (still positive, still

meaningful if rerun under the corrected readout) and two others fall below 82.21. We do not claim the methods themselves are broken; we claim that the published numbers cannot be read at face value without a verified token-readout.

The nine inference-time corrections we ran on the corrected baseline did not improve adversarial F1. The per-signal analysis in Section 5.3 indicates that the visual-grounding, image-noun, and attention-concentration signals these methods assume do not discriminate the cases the adversarial split is designed to query. Improving over the corrected adversarial baseline may require training-time changes that improve the model’s discrimination between visually co-occurring objects, rather than decoding-time changes alone. The correction itself is a one-line change in the evaluation code. We encourage future POPE-based work to verify the first-token generation behaviour of their specific model-template combination before committing to a fixed read-out token set.

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Justification: Theorem 1 (single-token readout equivalence) and Corollary 1 (failure mode of the two-token readout) in Section 6 state all assumptions explicitly and provide complete proofs. The hypothesis $t^* \in Y \cup N$ is load-bearing, not a formality: the InstructBLIP-Vicuna-7B case in Section 8 demonstrates a real violation in practice, where the parsed span did not isolate the answer token and the string-parse readout collapsed to all-yes while the single-token logit readout was unaffected.

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Justification: The evaluation script, diagnostic tool, and all 9,000 prediction records (predictions, labels, token IDs, and logits for all 16 yes/no token IDs) are released at the GitHub repository listed in Section 1. POPE uses COCO images, which are publicly available under the COCO Terms of Use.

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Justification: Section 3 reports the GPU (RTX 3070 Ti, 8 GB VRAM), quantization method, and total evaluation time (76.3 minutes for 9,000 questions). Section 7 compares per-question times for logit-based versus string-based evaluation.

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Question: Are new assets introduced in the paper clearly documented and is the documentation provided alongside the assets?

758 Answer: [Yes]
 759 Justification: The evaluation script and diagnostic tool are documented in the repository
 760 README with example commands (see Appendix B).

761 14. Crowdsourcing and Research with Human Subjects

762 Question: For crowdsourcing experiments and research with human subjects, does the paper
 763 include the working instructions, a description of the compensation paid to workers, and the
 764 number of annotators?

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766 Justification: No human subjects or crowdsourcing was involved.

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769 Question: Does the paper describe potential risks incurred by study participants as well
 770 as how these risks were mitigated, and whether and how IRB approval (or an equivalent
 771 approval/review based on the requirements of your country or institution) was obtained?

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774 A Token ID Reference and Tokenizer Traces

775 Table 9 lists all eight yes/no token IDs. The tokenizer decode outputs below confirm what each
 776 ID produces; these were run directly in Python against the LLaMA tokenizer loaded from the
 777 LLaVA-1.5-7B repository:

```
778 from transformers import AutoTokenizer
779 tok = AutoTokenizer.from_pretrained("llava-hf/llava-1.5-7b-hf")
780
781 tok.decode([3582]) # -> 'yes'
782 tok.decode([8241]) # -> 'Yes'
783 tok.decode([4874]) # -> ' yes' (space-prefixed)
784 tok.decode([3869]) # -> ' Yes' (space-prefixed) ***generated***
785 tok.decode([1217]) # -> 'no'
786 tok.decode([3782]) # -> 'No'
787 tok.decode([694]) # -> ' no' (space-prefixed)
788 tok.decode([1939]) # -> ' No' (space-prefixed) ***generated***
789
```

791 The two IDs marked ***generated*** are the only ones that appeared as first tokens across all
 792 9,000 questions in our runs.

Table 9: Yes/no token IDs for LLaMA-family models. *: the only IDs generated by LLaVA-1.5-7B after the ASSISTANT: prompt in our experiments.

Answer	Token ID	Decoded surface form
yes	3582	yes
yes	8241	Yes
yes	4874	' yes'
yes	3869*	' Yes'
no	1217	no
no	3782	No
no	694	' no'
no	1939*	' No'

B Diagnostic Script

```
python src/run_pope_2tok_baseline.py --split adversarial
```

The output JSON includes a `top_tokens_generated` field that sorts first-token IDs by frequency. Any ID appearing there that is absent from the active read-out token set must be added before the rule can be trusted on that model-template combination.

C Where the Mass Goes: Confusion Matrices and Yes-Rate Convergence

This appendix expands Section 3.4 with the visualisations that did not fit in the main body. Figure 10 contrasts the full 2×2 confusion matrices for adversarial POPE under the two protocols. Under eight-token the matrix is close to balanced (TP 1,174 vs. TN 1,318, FP 182, FN 326). Under two-token the false-positive cell jumps to 1,106; 924 of those originate from the $\rho=0.5621$ noise flips on true-no generations.

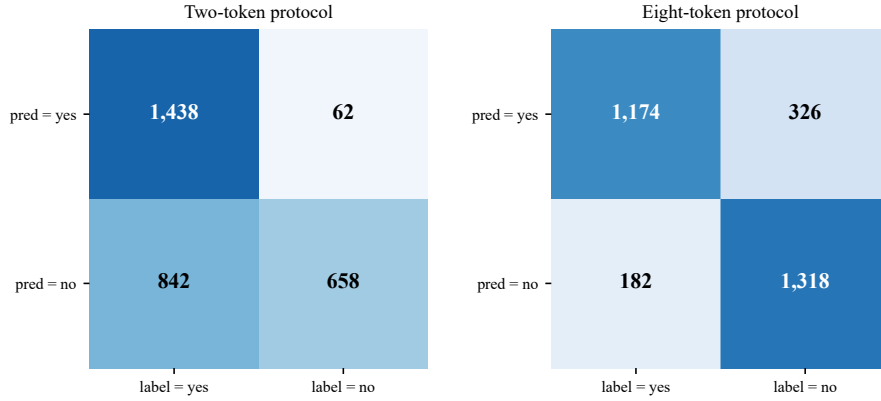


Figure 10: Confusion matrices for POPE adversarial (3,000 questions) under each protocol. The model’s underlying behaviour does not change between the two matrices; only the read-out does.

Figure 11 shows how the running yes-rate of each protocol evolves over the 3,000-question adversarial stream. The eight-token line stabilises near the ground-truth 0.500; the two-token line locks at 0.760 because the noise floor systematically biases the comparison toward yes.

Figure 12 shows the same yes-rate evolution for all three splits under both protocols on a single axis. The eight-token protocol gives yes-rates close to 0.5 on every split (adversarial 0.452, popular 0.434, random 0.411); the two-token protocol pushes every split toward 0.7+ regardless of the underlying question difficulty.

Popular and random splits. The same closed form in Eq. 3 predicts the gaps on the popular and random splits. Using the empirical (T, G) from Table 1 and the corresponding empirical TPs / FPs from our prediction logs:

$$\begin{aligned} \text{popular: } F1_8 &= 0.8498, \quad F1_2 = 0.7961, \quad \Delta = 0.0537, \\ \text{random: } F1_8 &= 0.8713, \quad F1_2 = 0.8397, \quad \Delta = 0.0316. \end{aligned}$$

The random-split gap is smallest because the model already says yes on proportionally more questions ($T/(T+G) = 0.411$), leaving fewer true-no generations to be flipped.

D Per-Split Confusion Matrices

Table 10 reports full 2×2 matrices under the eight-token rule. The adversarial false-positive rate (FP per negative question = $182/1500 = 12.1\%$) is roughly $4.2\times$ that of the random split ($43/1500 = 2.9\%$), consistent with the co-occurrence design of the adversarial negatives. All matrix cells are computed directly from the per-question prediction logs in `experiments/pope_{split}_2tok_vs_8tok.json`.

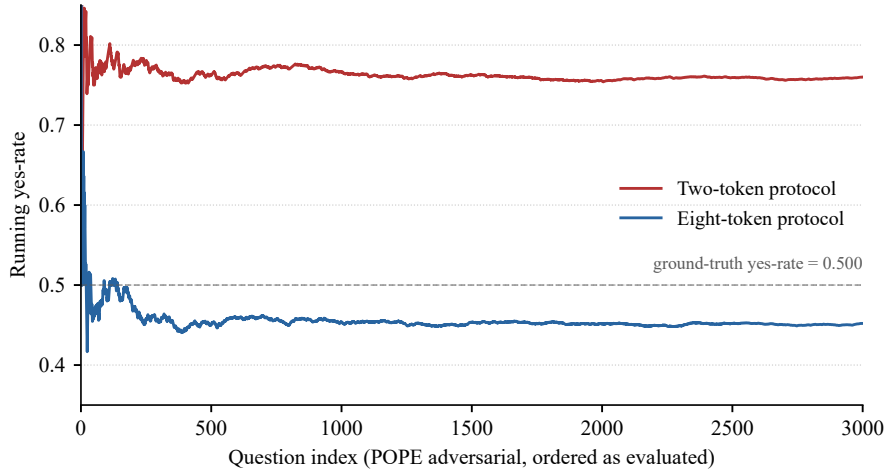


Figure 11: Running yes-rate over the 3,000-question POPE adversarial stream. The eight-token curve (blue) sits near the ground-truth yes-rate (dashed at 0.5); the two-token curve (red) sits at 0.76 because 56% of true-no generations get misread.

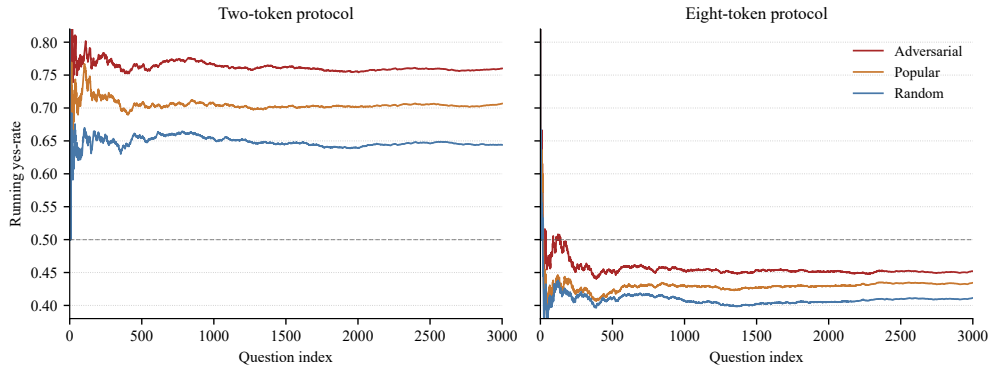


Figure 12: Running yes-rate across all three POPE splits under both protocols. The two-token protocol (left) produces a uniform yes-bias that is largely insensitive to which split the question came from. The eight-token protocol (right) reflects the underlying ground-truth mix.

822 E Prompt-Template Robustness on LLaVA-1.6-Mistral-7B

823 The master table in Section 8 fixes one prompt template per model. To check that
 824 the dynamic readout is stable under a change of template, rather than an artifact of
 825 the LLaVA-1.5 string, we ran a focused comparison on the first 500 POPE adversar-
 826 ial questions of LLaVA-1.6-Mistral under two templates: the LLaVA-1.5 paper template
 827 (USER: <image>\n{q} Answer yes or no only.\nASSISTANT:) and the model’s native chat
 828 template via processor.apply_chat_template. Across all 500 questions in each run, the only
 829 first tokens generated were id 5592 (Yes) and id 1770 (No); the LLaMA-2 IDs from the main paper
 830 never appeared. Table 11 reports the four readouts described in Section 7; the same numbers are
 831 plotted in Figure 13 for a direct visual comparison.

832 Three observations follow from Table 11, and all three are limited to the LLaVA-1.6-Mistral-7B /
 833 500-question setting we tested.

834 **The dynamic single-token and string-parse readouts agree exactly.** Under either prompt template
 835 the dynamic single-token readout and the string-parse readout report identical F1, precision, recall,
 836 and yes-rate to four decimals. The agreement is expected under greedy decoding: the dynamic
 837 single-token readout compares the max of {5081, 5592} to the max of {708, 1770}, and a greedy

Table 10: Confusion matrices under eight-token evaluation, full 3,000-question splits. Counts are exact, taken from the prediction JSON files; row sums equal 3,000 for each split.

Split	TP	TN	FP	FN
Adversarial	1,174	1,318	182	326
Popular	1,191	1,388	112	309
Random	1,191	1,457	43	309

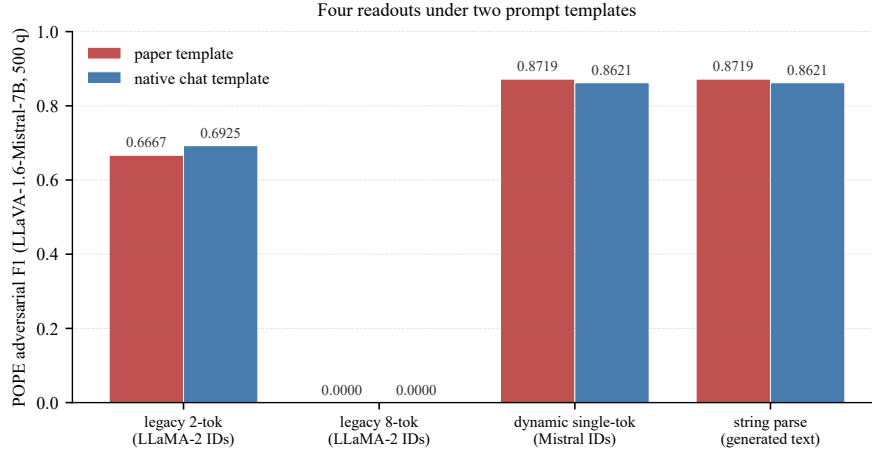


Figure 13: POPE adversarial F1 for LLaVA-1.6-Mistral-7B on the first 500 questions, under four readouts and two prompt templates. The dynamic single-token readout and the string-parse readout agree to four decimals within each prompt mode; the legacy LLaVA-1.5 eight-token readout collapses to zero because the LLaMA-2 IDs are never the argmax token on a Mistral-vocabulary model.

838 decoder emits whichever single token has the largest logit, which the string-parse readout then sees
839 verbatim. The exact match is useful as an internal consistency check: a discrepancy between the two
840 would signal either a multi-token surface form leaking through the generation (none did) or an error
841 in one of the implementations.

842 **Legacy LLaVA-1.5 readouts produce non-results on this vocabulary.** The legacy eight-token
843 readout has F1 = 0 under both prompt templates: none of the LLaMA-2 IDs 3582, 8241, 4874, 3869
844 or 1217, 3782, 694, 1939 are ever the argmax token, so the comparison between two zero-mass max-
845 logits resolves the same way on every question (here, toward no), and every question is labelled no.
846 The legacy two-token readout produces an all-yes or near-all-yes bias for the same reason: comparing
847 two never-generated IDs amounts to comparing two noise-floor values. This is the cross-model
848 analogue of the LLaVA-1.5 finding in the main paper, but with the failure mode larger in magnitude
849 because the LLaMA-2 IDs do not belong to the Mistral vocabulary at all rather than merely being
850 out-of-template.

851 **Absolute F1 is not directly comparable to LLaVA-1.5.** The 500-question slice, the different
852 vocabulary, the different prompt template, and the different model architecture all change the ab-
853 solute number. We do not claim 0.8719 or 0.8621 corrects any published LLaVA-1.6 baseline.
854 The 0.8719 here is the dynamic readout on the first 500 adversarial questions; on the full 3,000-
855 question adversarial split (Table 6) the same readout gives 0.8521, the small difference reflecting
856 the larger sample. The point of the cross-model run is narrower: to show that the four-readout
857 framework derived for LLaVA-1.5 ports cleanly to a second model when the token sets are re-
858 derived from that model’s tokenizer, and to show that transferring the LLaVA-1.5 IDs literally
859 (as the legacy rows do) is unsafe. Both artifacts (*_paper_template_token_audit.json and
860 *_native_chat_template_token_audit.json) are in the repository for inspection.

Table 11: Four readouts on the first 500 POPE adversarial questions under LLaVA-1.6-Mistral-7B. *legacy_2tok* and *legacy_8tok* use the LLaMA-2 IDs from the main paper and are reported only to illustrate the failure mode of an out-of-vocabulary readout. *dynamic_single* compares max-logits over the tokenizer-derived single-token IDs {5081, 5592} vs {708, 1770}. *string_parse* runs greedy decoding for up to 6 tokens and parses the first yes/no substring.

Prompt template	Readout	F1	Precision	Recall	Yes-rate
Paper template	legacy_2tok	0.6667	0.5000	1.0000	1.000
	legacy_8tok	0.0000	0.0000	0.0000	0.000
	dynamic_single	0.8719	0.8352	0.9120	0.546
	string_parse	0.8719	0.8352	0.9120	0.546
Native chat template	legacy_2tok	0.6925	0.5297	1.0000	0.944
	legacy_8tok	0.0000	0.0000	0.0000	0.000
	dynamic_single	0.8621	0.9517	0.7880	0.414
	string_parse	0.8621	0.9517	0.7880	0.414

F Survey Procedure and Community-Impact Denominators

This appendix expands Section 9 with the procedure for filling the survey scaffold (Table 8) and the full definitions of the two denominators and the numerator that the main body reports.

For each candidate paper, the survey procedure is:

1. Locate the LLaVA-1.5-7B POPE adversarial baseline F1 in the paper’s main results table.
2. If the paper’s released code contains a yes/no readout step, check whether it uses logit indices (and which) or a string match.
3. If the readout is not stated and not in the released code, mark the row as *readout: not stated* and the conclusion as I.
4. If the readout uses logit indices that include the LLaMA-2 IDs 3582 and 1217 only, mark conclusion R; if it uses string matching, mark the conclusion as “not necessarily affected” (NA in the conclusion column).

The CSV in `references/paper_survey.csv` mirrors Table 8 and is the canonical place to add rows; the build process re-renders the table from the CSV on request.

We deliberately do not headline a number of the form “ N papers’ gains evaporate”. The honest framing has two denominators and one numerator:

- **Denominator 1 (verifiable):** the number of papers in the survey whose readout we can verify either by reading the released code or by direct statement in the paper.
- **Denominator 2 (broader):** the number of papers we have surveyed in total, including those marked I.
- **Numerator:** among Denominator 1, the number whose reported gain shrinks or reverses sign when measured against the eight-token baseline of 82.21.

Reporting Numerator / Denominator 1 gives a conservative lower bound on the share of affected published gains; reporting the size of Denominator 2 minus Denominator 1 gives an explicit measure of how much of the literature remains indeterminate. As of the present version, Denominator 1 = 3, Numerator = 3 (Tang et al. shrinks from +10.81 to +2.14, Seo et al. reverses to −2.21, Chen et al. reverses to −0.41), and Denominator 2 = 8. We invite community contributions to expand both denominators via pull requests against `references/paper_survey.csv`.